

DBMS PROJECT REPORT

REQUIREMENT ANALYSIS AND CUSTOMER DATABASE MODULE

E-Commerce Database System

1. Introduction

An e-commerce application is designed to support online purchasing by connecting customers, sellers, products, orders, and payment activities. A well-structured database is required to store these details and maintain the relationships between different activities involved in the purchasing process.

The Requirement Analysis and Customer Database Module focuses on understanding the requirements of an e-commerce system and designing the basic database structure needed to manage customer information.

The module identifies the major entities involved in the system and explains their attributes and relationships. It also demonstrates the creation, insertion, viewing, and updating of customer records in the database.

2. Requirement Analysis

Understanding the E-Commerce Business

The e-commerce application allows customers to purchase products through an online platform. The system connects different activities such as product browsing, cart management, order placement, payment, delivery, reviews, and customer support.

A database plays an important role in storing the information generated during these activities. It maintains customer details, product information, seller information, orders, payments, and reviews while maintaining relationships between them.

The main purpose of requirement analysis is to identify the information that needs to be stored and understand how the different entities interact with each other.

3. Process Flow of the E-Commerce System

Browse / Search Product

Customers can explore the products available on the e-commerce platform. They can search for a particular product based on their requirements.

Add to Cart

After selecting a suitable product, the customer can add the product to their shopping cart.

Order Placed

The customer provides the required delivery information and confirms the order.

Checkout / Payment

The order moves to the checkout stage. The customer completes the payment using one of the available payment methods.

Delivery

After successful payment, the order is processed for dispatch and delivered to the address provided by the customer.

Review & Support

After receiving the product, the customer can provide feedback through a review. The customer can also request assistance for returns, replacements, installation, or other product-related issues.

4. Entities Identified in the System

The major entities identified for the e-commerce database are:

1. Customer
2. Seller
3. Product
4. Order
5. Payment
6. Review

Customer Entity

The Customer entity stores the basic information of users who purchase products through the e-commerce platform. Each customer is identified using a unique Customer_ID, which acts as the primary key.

Key Attributes:

Attribute	Description
Customer_ID (PK)	Unique identification number of the customer
Customer_Name	Name of the customer
Email	Email address of the customer
Phone	Contact number of the customer
created_at	Date and time when the customer record was created

Key Relationship: A customer can place multiple orders and can also provide multiple reviews. Therefore, the Customer entity has a One-to-Many relationship with Order and Review.

Seller Entity

The Seller entity stores information about vendors who provide products through the e-commerce platform. A seller can provide multiple products.

Key Attributes:

Attribute	Description
seller_id (PK)	Unique ID of the seller
company_name	Name of the seller/company
email	Seller's email address
phone	Seller's contact number
rating	Rating of the seller
bank_details	Seller's banking information

Key Relationship: A seller can provide multiple products. Therefore, the Seller entity has a One-to-Many relationship with Product.

Product Entity

The Product entity represents the products listed by sellers for customers to purchase. It stores information such as product name, description, price, available stock, and category-related information.

Key Attributes:

Attribute	Description
product_id (PK)	Unique ID of the product
seller_id (FK)	Identifies the seller providing the product
product_name	Name of the product
description	Description of the product
price	Price of the product
stock_quantity	Available quantity of the product

Key Relationship: A product belongs to one seller, creating a Many-to-One relationship with Seller. A product can also receive multiple reviews, creating a One-to-Many relationship with Review.

Order Entity

The Order entity maintains information about purchases made by customers. It records details such as the order date, order status, total amount, and delivery address.

Key Attributes:

Attribute	Description
order_id (PK)	Unique ID of the order
customer_id (FK)	Identifies the customer who placed the order
order_date	Date on which the order was placed
status	Current status of the order
total_amount	Total amount of the order
shipping_address	Delivery address

Key Relationship: The Order entity has a Many-to-One relationship with Customer, a One-to-One relationship with Payment, and a Many-to-Many relationship with Product through OrderItem.

Payment Entity

The Payment entity records the transaction information associated with an order. It maintains information about the payment method, payment status, transaction ID, amount, and payment date.

Key Attributes:

Attribute	Description
payment_id (PK)	Unique ID of the payment
order_id (FK)	Identifies the related order
payment_method	Method used for payment
status	Payment status
transaction_id	Unique transaction reference
amount	Amount paid
payment_date	Date of payment

Key Relationship: The Payment entity has a One-to-One relationship with Order.

Review Entity

The Review entity maintains feedback provided by customers for products. It stores the customer's rating and comments after experiencing the product.

Key Attributes:

Attribute	Description
review_id (PK)	Unique ID of the review
customer_id (FK)	Identifies the customer
product_id (FK)	Identifies the reviewed product
rating	Rating given by the customer
comment	Customer's feedback
created_at	Date and time when the review was created

Key Relationship: The Review entity has a Many-to-One relationship with Customer and Product.

5. Customer Database Module

The Customer Database Module is responsible for maintaining the information of customers using the e-commerce application.

The Customer table stores the customer's basic personal and registration information. The Customer_ID acts as the unique identifier for every customer.

The module mainly demonstrates the following database operations:

- Create customer records
- Insert customer information
- View stored customer information
- Update existing customer information

6. Customer Table Creation

The Customer table is created to store customer information in the database. The table contains the following columns:

Column Name	Purpose
Customer_ID	Unique identifier for each customer
Customer_Name	Stores customer name
Email	Stores customer email
Phone	Stores customer phone number
created_at	Stores customer registration date/time

The Customer_ID is used as the primary key so that each customer can be uniquely identified.

```
SQL> CREATE TABLE CUSTOMER (CUSTOMER int PRIMARY KEY, CUSTOMER_NAME VARCHAR(100) NOT NULL, EMAIL VARCHAR(100) UNIQUE, PHONE VARCHAR(15) UNIQUE, ADDRESS VARCHAR(200), REGISTRATION_DATE DATE NOT NULL);
```

Table created.

contains values corresponding to the columns defined in the Customer table. This operation allows the database to store information about multiple customers.

```
SQL> INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID', '&CUSTOMER_NAME', '&EMAIL', &PHONE, '&ADDRESS', TO_DATE('&DATE', 'YYYY-MM-DD'));
Enter value for customer_id: 101
Enter value for customer_name: SNEHA R
Enter value for email: Sneha@gmail.com
Enter value for phone: 9876543215
Enter value for address: CHENNAI
Enter value for date: 2026-02-01
old 1: INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID', '&CUSTOMER_NAME', '&EMAIL', &PHONE, '&ADDRESS', TO_DATE('&DATE', 'YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('101', 'SNEHA R', 'Sneha@gmail.com', 9876543215, 'CHENNAI', TO_DATE('2026-02-01', 'YYYY-MM-DD'))

1 row created.
```

```
SQL> /
Enter value for customer_id: VIJAY KUMAR
Enter value for customer_name: /
Enter value for email:
Enter value for phone:
Enter value for address:
Enter value for date:
old 1: INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID', '&CUSTOMER_NAME', '&EMAIL', &PHONE, '&ADDRESS', TO_DATE('&DATE', 'YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('VIJAY KUMAR', '/', '', '', TO_DATE('', 'YYYY-MM-DD'))
INSERT INTO CUSTOMER VALUES('VIJAY KUMAR', '/', '', '', TO_DATE('', 'YYYY-MM-DD'))
*
```

```
ERROR at line 1:
ORA-00936: missing expression
```

```
SQL> /
Enter value for customer_id: 102
Enter value for customer_name: VIJAY KUMAR
Enter value for email: Vijay@gmail.com
Enter value for phone: 9876543216
Enter value for address: TIRUNELVELI
Enter value for date: 2026-02-05
old 1: INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID', '&CUSTOMER_NAME', '&EMAIL', &PHONE, '&ADDRESS', TO_DATE('&DATE', 'YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('102', 'VIJAY KUMAR', 'Vijay@gmail.com', 9876543216, 'TIRUNELVELI', TO_DATE('2026-02-05', 'YYYY-MM-DD'))

1 row created.
```

```
SQL> /
Enter value for customer_id: 103
Enter value for customer_name: ANITHA P
Enter value for email: Anitha@gmail.com
Enter value for phone: 9876543217
Enter value for address: ERODE
```

```

SQL> /
Enter value for customer_id: 104
Enter value for customer_name: SURESH B
Enter value for email: Suresh@gmail.com
Enter value for phone: 9876543218
Enter value for address: SALEM
Enter value for date: 2026-02-15
old 1: INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID','&CUSTOMER_NAME','&EMAIL','&PHONE','&ADDRESS',TO_DATE('&DATE','YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('104','SURESH B','Suresh@gmail.com',9876543218,'SALEM',TO_DATE('2026-02-15','YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for customer_id: 105
Enter value for customer_name: KAMESH
Enter value for email: Kamesh@gmail.com
Enter value for phone: 9840371453
Enter value for address: KIRABAKKAM
Enter value for date: 2026-02-20
old 1: INSERT INTO CUSTOMER VALUES('&CUSTOMER_ID','&CUSTOMER_NAME','&EMAIL','&PHONE','&ADDRESS',TO_DATE('&DATE','YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('105','KAMESH','Kamesh@gmail.com',9840371453,'KIRABAKKAM',TO_DATE('2026-02-20','YYYY-MM-DD'))

1 row created.

```

After inserting the required records, the Customer table can be displayed to verify whether the information has been successfully stored in the database.

```

SQL> SELECT * FROM CUSTOMER;

```

CUSTOMER_ID	CUSTOMER_NAM	EMAIL	PHONE	ADDRESS	REGISTERAT
101	SNEHA R	Sneha@gmail.com	9876543215	CHENNAI	01-FEB-26
102	VIJAY KUMAR	Vijay@gmail.com	9876543216	TIRUNELVELI	05-FEB-26
103	ANITHA P	Anitha@gmail.com	9876543217	ERODE	10-FEB-26
104	SURESH B	Suresh@gmail.com	9876543218	SALEM	15-FEB-26
105	KAMESH	Kamesh@gmail.com	9840371453	KIRABAKKAM	20-FEB-26

8. Updating Customer Information

The UPDATE statement is used to modify an existing customer's information. A particular customer can be selected using the customer_id, and the required field can then be changed.

This query changes the phone number of the customer whose Customer_ID is 103. After performing the update, the table can be displayed again to verify the modification.

```

SQL> UPDATE CUSTOMER SET PHONE = 9876543212 WHERE CUSTOMER_ID = '103';

1 row updated.

```

9. Conclusion

The Requirement Analysis and Customer Database Module establishes the basic structure required for managing customer information in an e-commerce system.

The requirement analysis identifies the major entities such as Customer, Seller, Product, Order, Payment, and Review, along with their attributes and relationships.

The Customer Database Module demonstrates how customer records can be created, inserted, stored, viewed, and updated in the database.

Overall, the identified entities and their relationships provide a foundation for connecting customers with orders, products, payments, and reviews, which can be further developed into a complete e-commerce database system.